TCS NQT PYQ QUESTIONS

Company: TCS NQT
Round: Online
Year: 2025
Difficulty: High
Topic: Algebra (Symmetric Expressions)

Problem Description:

If x is a positive real number satisfying $x + 1/x = 5$, find the value of $x^5 + 1/x^5$ (that is, x to the power 5 plus 1 over x to the power 5).

Options:

A) 2500
B) 2515
C) 2525
D) 2530

Correct Answer:

C) 2525

Approach to Solve This Problem:

STEP 1:

Build up the powers step by step using the identity $(a+b)^2 = a^2+b^2+2ab$ and similar expansions, with $a = x$ and $b = 1/x$ (so $ab = 1$).

$x^2 + 1/x^2 = (x + 1/x)^2 - 2 = 5^2 - 2 = 23$.

STEP 2:

$x^3 + 1/x^3 = (x + 1/x)^3 - 3(x + 1/x) = 125 - 15 = 110$.

$x^4 + 1/x^4 = (x^2 + 1/x^2)^2 - 2 = 23^2 - 2 = 527$.

STEP 3:

Now use: $x^5 + 1/x^5 = (x + 1/x)(x^4 + 1/x^4) - (x^3 + 1/x^3)$

$= (5 \times 527) - 110 = 2635 - 110 = 2525$.

CORRECT ANS = 2525